

1.

A 10-mm square scribed on a large steel pressure vessel. After pressurization, a biaxial stress condition developed, with $\sigma_x = 80$ MPa and $\sigma_y = 40$ MPa.

$E=200$ GPa, $\nu=0.3$

Determine the change in x dimension δ_x in micrometers

2.

What does Polar Moment of Inertia (denoted by J) physically represent in torsion?

- A. The resistance of a material to axial loading
- B. The resistance of a shaft to bending deformation
- C. The resistance of a shaft's cross section to twisting under applied torque
- D. The energy stored in a shaft during elastic deformation

3.

The angle of twist ϕ (in radians) for a cylindrical shaft of length L , under torque T , is given by

- A. TL/JG
- B. MC/I
- C. VQ/It
- D. PL/AE
- E. Tc/J

4.

In the elastic torsion equation T_c/J , what does the symbol J stand for?

- A. Young's modulus
- B. Moment of Inertia about the neutral axis
- C. Polar Moment of Inertia
- D. Poisson's ratio

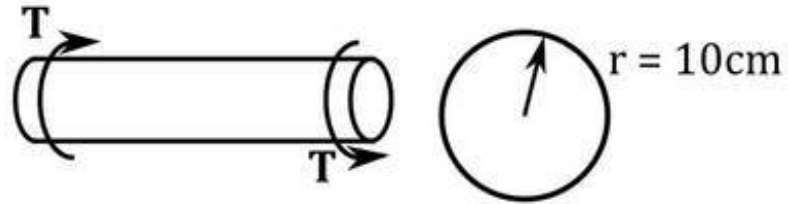
5.

We have a brittle cylindrical shaft, subjected to equal and opposite torques along opposite ends. The brittle fracture will occur along the following plane :

- A. Perpendicular to the axis of the shaft
- B. parallel to the axis of the shaft
- C. along a plane 45° to the axis of the shaft

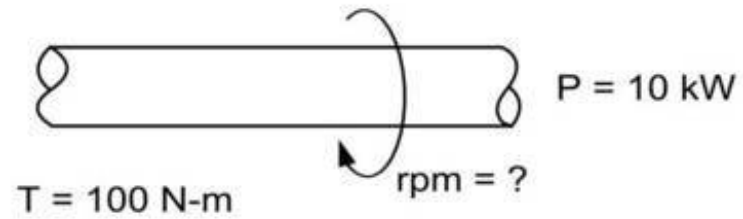
6.

A cylinder of radius $r = 10$ cm is subject to a torque T . At what radius in the shaft will the shear stress due to torsion be 75% of the maximum stress due to torsion ? (Submit your answer in units of cm.)



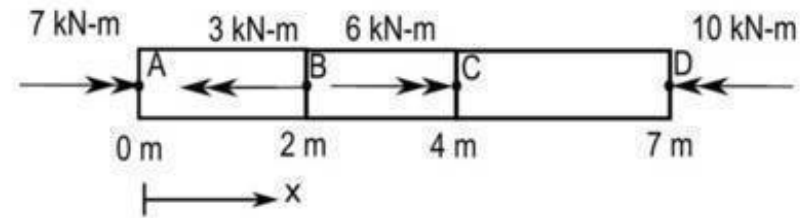
7.

A shaft reaches its stress limit when the applied torque is 100 N-m. If this shaft is to be used to transfer 10 kW of power, at what speed must it rotate in rpm ?



8.

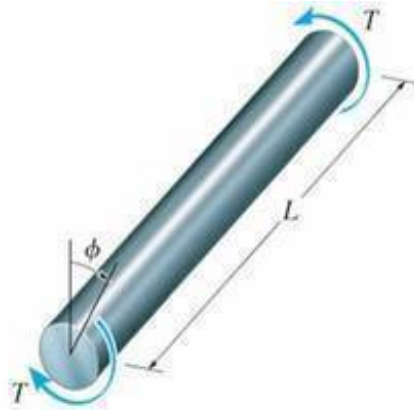
Consider a uniform cylinder with torques applied as shown. With respect to the left end, i.e. deflection at left end is zero, at which point (B, C, or D,) will the deflection of this member be the greatest ?



- A. A
- B. B
- C. C
- D. D

9.

Determine the deflection ϕ (in radians) over the length of the cylinder when it is subject to a torque $T = 750$ N-m. The cylinder has properties as listed with the figure.



length $L = 10$ m

Radius $r = 24$ mm

Area $A = 1.77 \times 10^{-3} \text{ m}^2$

Poissons Ratio $\nu = 0.3$

Modulus of Elasticity $E = 200$ GPa

Polar Moment of Inertia $J = 0.5 \times 10^{-6} \text{ m}^4$

Shear Modulus $G = 75$ GPa

10.

We have steel metal shafts with the same E and same length L and same cross sectional area, A . However, one of the shafts is hollow. Which shaft would experience a larger angle of twist for the same applied torque T ? Choose one of the answers but be prepared to prove your answer mathematically.

- A. Solid shaft
- B. Hollow shaft